
Practical 1

To solve partial differential equation

`DSolve[eqn, u[x,y], {x,y}]`- to solve a PDE for $u[x,y]$.

The general solutions to partial differential equations involve arbitrary functions. These functions are labeled as $C[i]$. `DSolve` works for PDEs having two independent variables.

First-Order PDEs - Linear and Quasi-Linear PDEs

A first-order PDE for an unknown function $u(x,y)$ is said to be linear if it can be expressed in the form

$$a(x, y) \frac{\partial u(x, y)}{\partial x} + b(x, y) \frac{\partial u(x, y)}{\partial y} + c(x, y) u(x, y) = d(x, y)$$

The PDE is said to be quasilinear if it can be expressed in the form

$$a(x, y, u(x, y)) \frac{\partial u(x, y)}{\partial x} + b(x, y, u(x, y)) \frac{\partial u(x, y)}{\partial y} = c(x, y, u(x, y))$$

Q1. Solve the first order linear partial differential equation

$$y u_x - x u_y = 0$$

Solution:

```
In[ ]:= eqn = y D[u[x, y], x] - x D[u[x, y], y] == 0;  
DSolve[eqn, u[x, y], {x, y}]
```

```
Out[ ]:= {{u[x, y] -> C[1] [1/2 (x^2 + y^2)]}}
```

Q2. Solve the first order linear partial differential equation

$$x u_x - y u_y = 0$$

Solution:

```
In[ ]:= eqn1 = x D[u[x, y], x] - y D[u[x, y], y] == 0;  
DSolve[eqn1, u[x, y], {x, y}]
```

```
Out[ ]:= {{u[x, y] -> C[1] [x y]}}
```

Q. Solve the first order linear partial differential equation

$$2 u_x + 3 u_y + u = 0$$

Solution:

```
In[ ]:= eqna = 2 D[u[x, y], x] + 3 D[u[x, y], y] + u[x, y] == 0;  
DSolve[eqna, u[x, y], {x, y}]
```

```
Out[ ]:= {{u[x, y] -> E^(-x/2) C[1] [-3 x/2 + y]}}
```

Second-Order PDEs

The general form of a linear second-order PDE is

$$a \frac{\partial^2 u}{\partial^2 x} + b \frac{\partial^2 u(x,y)}{\partial x \partial y} + c \frac{\partial^2 u}{\partial^2 y} + d \frac{\partial u}{\partial x} + e \frac{\partial u}{\partial y} + fu = g$$

Here $u=u(x,y)$, and a, b, c, d, e, f , and g are functions of x and y only—they do not depend on u . If $g=0$, the equation is said to be **homogeneous**. *DSolve* can find the general solution for a homogeneous linear second-order PDE of the form

$$a \frac{\partial^2 u}{\partial^2 x} + b \frac{\partial^2 u(x,y)}{\partial x \partial y} + c \frac{\partial^2 u}{\partial^2 y} = 0.$$

Here, a, b , and c are constants

Q3. Solve the second order linear partial differential equation

$$u_{yy} + u = 0$$

Solution:

```
In[ ]:= eqn2 = D[u[x, y], {y, 2}] + u[x, y] == 0;
DSolve[eqn2, u[x, y], {x, y}]
```

```
Out[ ]:= {{u[x, y] -> Cos[y] c1[x] + Sin[y] c2[x]}}
```

Q4. Solve the Laplace's equation

```
In[ ]:= LaplaceEqn = D[u[x, y], {x, 2}] + D[u[x, y], {y, 2}] == 0;
DSolve[LaplaceEqn, u[x, y], {x, y}]
```

```
Out[ ]:= {{u[x, y] -> c1[i x + y] + c2[-i x + y]}}
```

Q5. Solve the second order linear partial differential equation

$$3u_{xx} + u_{xy} + 5u_{yy} = 0$$

Solution:

```
In[ ]:= eqn3 = 3 D[u[x, y], {x, 2}] + D[u[x, y], x, y] + 5 D[u[x, y], {y, 2}] == 0;
DSolve[eqn3, u[x, y], {x, y}]
```

```
Out[ ]:= {{u[x, y] -> c1[1/6 (-1 + i sqrt(59)) x + y] + c2[1/6 (-1 - i sqrt(59)) x + y]}}
```

Q. Solve the second order linear partial differential equation

$$x^2 u_{xx} - y^2 u_{yy} = 0$$

Solution:

```
In[ ]:= DSolve[x^2 D[u[x, y], {x, 2}] - y^2 D[u[x, y], {y, 2}] == 0, u[x, y], {x, y}]
```

```
Out[ ]:= DSolve[-y^2 u^{(0,2)}[x, y] + x^2 u^{(2,0)}[x, y] == 0, u[x, y], {x, y}]
```

Practicals-2

Solution and plotting of Cauchy problem for first

order PDEs

Problem-:1 Obtain the solution of the linear equation

$$u_x - u_y = 1,$$

with the Cauchy data $u(x,0) = x^2$.

Plot the integral surface with in the range $\{x, -4, 4\}$ & $\{y, -5, 5\}$.

Solution-:

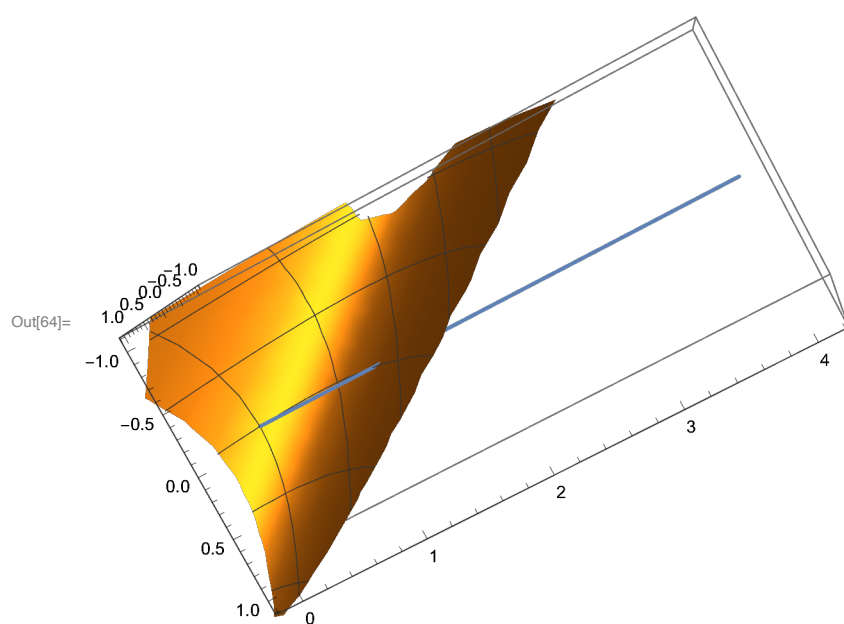
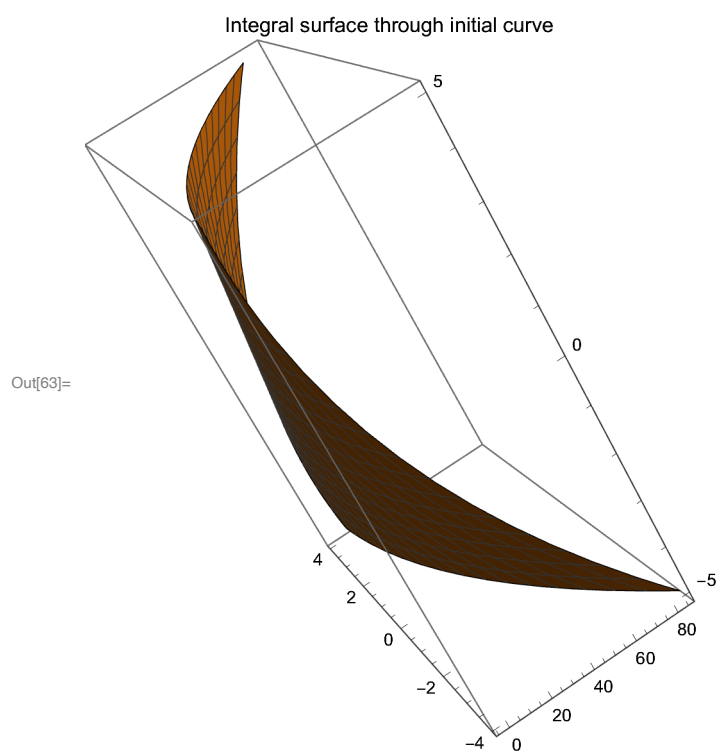
```

In[60]:= pde = D[u[x, y], x] - D[u[x, y], y] == 1
sol = DSolve[{pde, u[x, 0] == x^2}, u[x, y], {x, y}]
g1 = ParametricPlot3D[{0, t^2, 0}, {t, -2, 2}];
g2 = Plot3D[u[x, y] /. sol, {x, -4, 4}, {y, -5, 5},
  PlotLabel -> "Integral surface through initial curve"]
Show[g1, g2]

```

```
Out[60]=  $-u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1$ 
```

```
Out[61]=  $\left\{ \left\{ u[x, y] \rightarrow x^2 - y + 2xy + y^2 \right\} \right\}$ 
```



Problem-2: Find the solution of the equation

$$y u_x - 2 x y u_y = 2 x u,$$

with the Cauchy data $u(0,y) = y^3$.

Plot the integral surface with in the range $\{x, -7, 7\}$ & $\{y, -5, 5\}$.

Solution:-

$$h = y * D[u[x, y], x] - 2 * x * y * D[u[x, y], y] == 2 * x * u[x, y]$$

$$\text{sol3} = \text{DSolve}[\{h, u[0, y] == y^3\}, u[x, y], \{x, y\}]$$

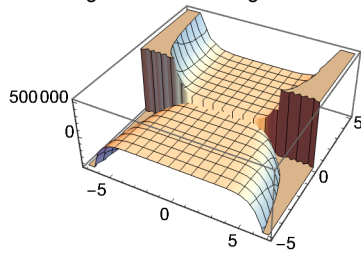
$$\text{Plot3D}[u[x, y] /. \text{sol3}, \{x, -7, 7\}, \{y, -5, 5\},$$

$$\text{PlotLabel} \rightarrow \text{"Integral surface through initial curve"}]$$

$$-2 x y u^{(0,1)}[x, y] + y u^{(1,0)}[x, y] == 2 x u[x, y]$$

$$\left\{ \left\{ u[x, y] \rightarrow \frac{(x^2 + y)^4}{y} \right\} \right\}$$

Integral surface through initial curve



Problem-:3 Determine the integral surfaces of the equation

$$u_x + u_y = u^2,$$

with the data $x+y = 0, u=1$.

Plot the integral surface with in the range $\{x, -10, 10\}$ & $\{y, -10, 10\}$.

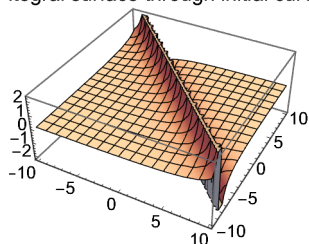
Solution:-

```
Eqn = D[u[x, y], x] + D[u[x, y], y] == u[x, y] * u[x, y]
DSolve[{Eqn, u[x, -x] == 1}, u[x, y], {x, y}]
Plot3D[u[x, y] /. %, {x, -10, 10}, {y, -10, 10},
  PlotLabel -> "Integral surface through initial curve"]
```

$$u^{(0,1)}[x, y] + u^{(1,0)}[x, y] = u[x, y]^2$$

$$\left\{ \left\{ u[x, y] \rightarrow -\frac{2}{-2 + x + y} \right\} \right\}$$

Integral surface through initial curve



Problem:-4 Obtain the solution of the linear equation

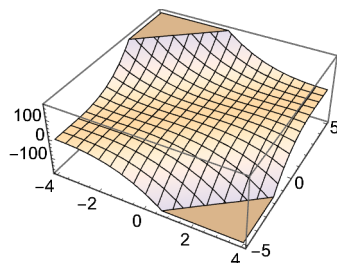
$$u_x + u_y = 1,$$

with the Cauchy data $u(x, 2x) = x^3$.

Plot the integral surface with in the range $\{x, -4, 4\}$ & $\{y, -5, 5\}$.

Solution:-

```
D[u[x, y], x] + D[u[x, y], y] == 1
sol = DSolve[
  {D[u[x, y], x] + D[u[x, y], y] == 1, u[x, 2 x] == x^3}, u[x, y], {x, y}]
Plot3D[u[x, y] /. sol, {x, -4, 4}, {y, -5, 5}]
u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1
{ {u[x, y] -> 2 x - x^3 - y + 3 x^2 y - 3 x y^2 + y^3} }
```



Problem-5: Find the solution of the equation

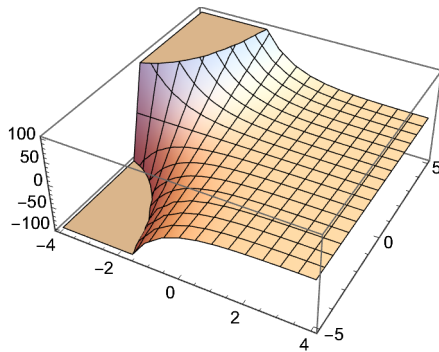
$$u_x + y * u_y = 0,$$

with the Cauchy data $u(0, y) = 4 * y$.

Plot the integral surface with in the range $\{x, -4, 4\}$ & $\{y, -5, 5\}$.

Solution:-

```
pde5 = D[u[x, y], x] + y * D[u[x, y], y] == 0
sol5 = DSolve[{pde5, u[0, y] == 4 y}, u[x, y], {x, y}]
Plot3D[u[x, y] /. sol5, {x, -4, 4}, {y, -5, 5}]
 $y u^{(0,1)}[x, y] + u^{(1,0)}[x, y] = 0$ 
 $\{ \{ u[x, y] \rightarrow 4 e^{-x} y \} \}$ 
```



Problem:-6 Determine the integral surfaces of the equation

$$u_x + u_y = u^2,$$

with the data $u(x,0)=\tanh(x)$.

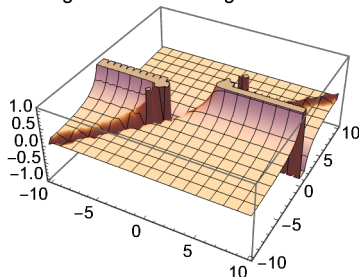
Plot the integral surface with in the range $\{x, -10, 10\}$ & $\{y, -10, 10\}$.

Solution:-

```
pde7 = D[u[x, y], x] + D[u[x, y], y] == u[x, y] * u[x, y]
sol7 = DSolve[{pde7, u[x, 0] == Tanh[x]}, u[x, y], {x, y}]
Plot3D[u[x, y] /. %, {x, -10, 10}, {y, -10, 10},
  PlotLabel -> "Integral surface through initial curve"]
 $u^{(0,1)}[x, y] + u^{(1,0)}[x, y] = u[x, y]^2$ 
```

$$\left\{ \left\{ u[x, y] \rightarrow \frac{1}{-y + \text{Coth}[x - y]} \right\} \right\}$$

Integral surface through initial curve



Problem:-7 Determine the integral surfaces of the equation

$$u_x + u_y = 5u^2,$$

with the data $u(x, 0) = x^2$.

Plot the integral surface with in the range $\{x, -4, 4\}$ & $\{y, -5, 5\}$.

Solution:-

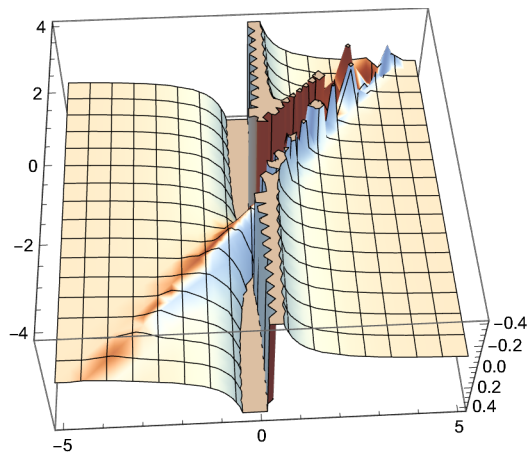
```
pde8 = D[u[x, y], x] + D[u[x, y], y] == 5 * u[x, y] * u[x, y]
```

```
sol8 = DSolve[{pde8, u[x, 0] == x^2}, u[x, y], {x, y}]
```

```
Plot3D[u[x, y] /. sol8, {x, -4, 4}, {y, -5, 5}]
```

$$u^{(0,1)}[x, y] + u^{(1,0)}[x, y] = 5 u[x, y]^2$$

$$\left\{ \left\{ u[x, y] \rightarrow -\frac{(x-y)^2}{-1 + 5x^2y - 10xy^2 + 5y^3} \right\} \right\}$$



Practical 2

Plotting the Characteristics for the first order PDE

Q1. Plot the Characteristic curves of the equation: $(u - y) u_x + y u_y = x + y$
(1)

Solution: The characteristic equations are:

$$\frac{dx}{u-y} = \frac{dy}{y} = \frac{du}{x+y}$$

On taking I + III = II, we get

$$(u+x)/y = C_1$$

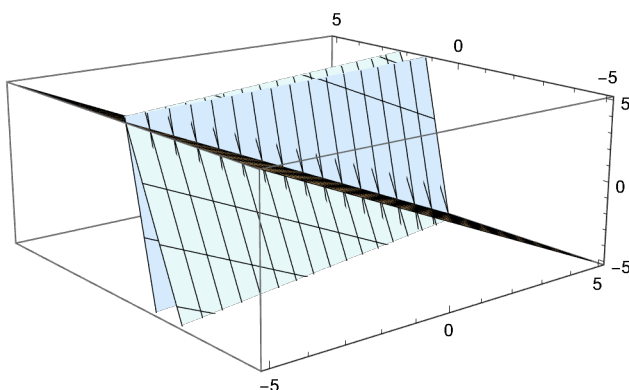
On taking I + II = III, we get

$$(x+y)^2 - u^2 = C_2$$

where C_1 and C_2 are constants.

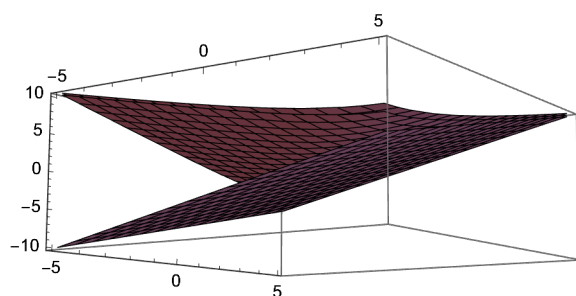
We plot the characteristic curves of the equation (1) for C_1 and C_2 equal to 0, 5, 10.

```
f0 = Plot3D[-x, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];  
f1 = Plot3D[5 * y - x, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];  
f2 = Plot3D[10 * y - x, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];  
g1 = Show[f0, f1, f2]
```



```
h0 = Plot3D[x + y, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];  
h1 = Plot3D[Sqrt[(x + y)^2 + 5], {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];  
h2 = Plot3D[Sqrt[(x + y)^2 + 10], {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];
```

```
g2 = Show[h0, h1, h2]
```



Q2. Plot the Characteristic curves of the equation : $xu_x + yu_y = u \dots \dots (2)$

Solution : The characteristic equations are :

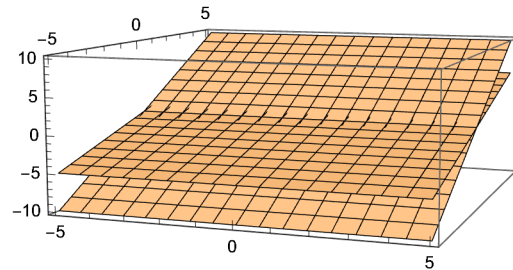
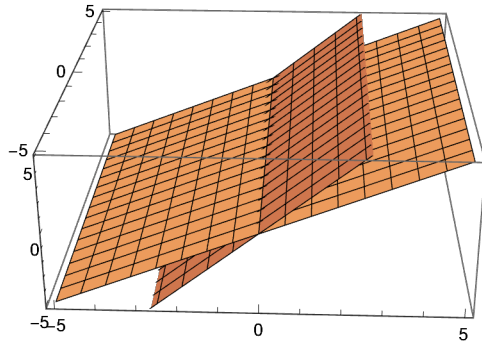
$$\frac{dx}{x} = \frac{dy}{y} = \frac{du}{u}$$

From these equations we get: $x/u = C_1$ and $y/u = C_2$ where C_1 and C_2 are constants
We plot the characteristic curves of (2) by taking C_1 and C_2 as 1 and 2.

```
f0 = Plot3D[x, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];
f1 = Plot3D[2 x, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];
h0 = Plot3D[2 y, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];
h1 = Plot3D[y, {x, -5, 5}, {y, -5, 5}, PlotPoints -> 10];
g1 = Show[f0, f1];
```

```
g2 = Show[h0, h1];
```

```
Show[GraphicsArray[{g1, g2}]]
```



**Q3. Plot the Characteristic curves of the equation : $x^2 u_x + y^2 u_y =$
 $(x + y) u \dots \dots (3)$**

Solution : The characteristic equations are :

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{du}{(x+y)u}$$

From the first of these equations , we get: $x^{-1} - y^{-1} = C_1$,(i)

where C_1 is an arbitrary constant.

Also, $\frac{dx-dy}{x^2-y^2} = \frac{du}{(x+y)u}$

or

$$\frac{d(x-y)}{x-y} = \frac{du}{u}.$$

This gives

$$\frac{x-y}{u} = C_2, \text{(ii)}$$

where C_2 is constant.

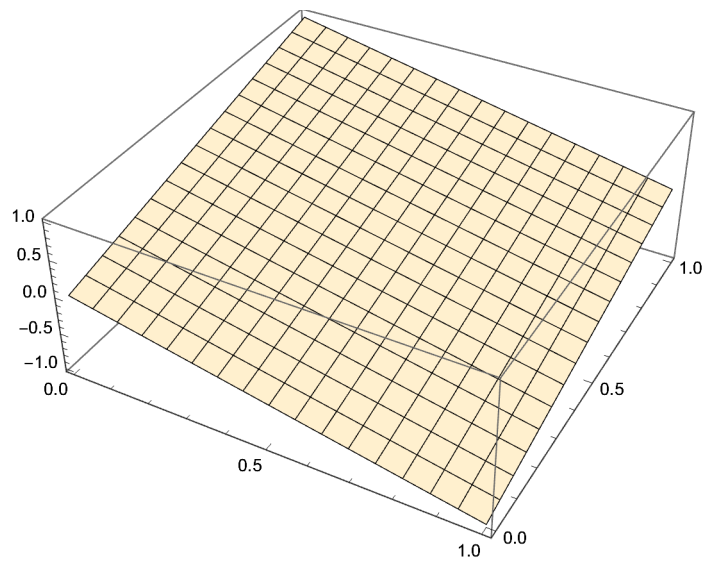
Further (i) and (ii) give

$$\frac{xy}{u} = C_3, \text{(iii)}$$

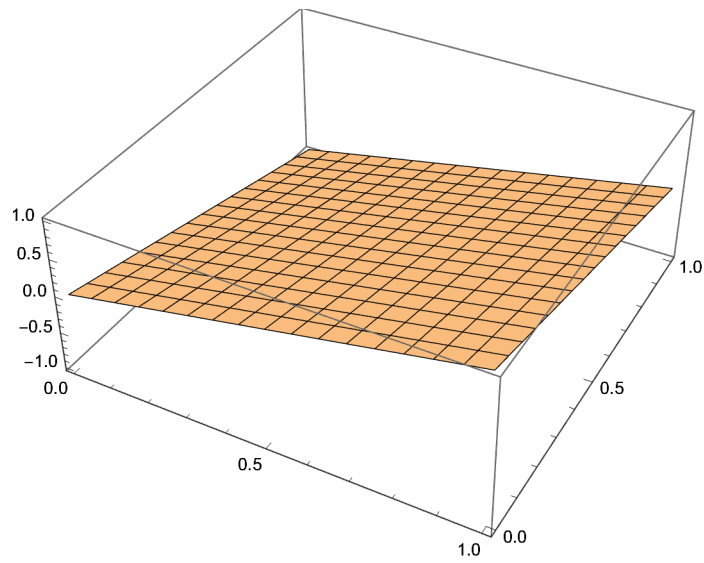
where C_3 is a constant.

We plot the characteristic curves (ii) and (iii) by taking C_2 and C_3 as -1,1,2.

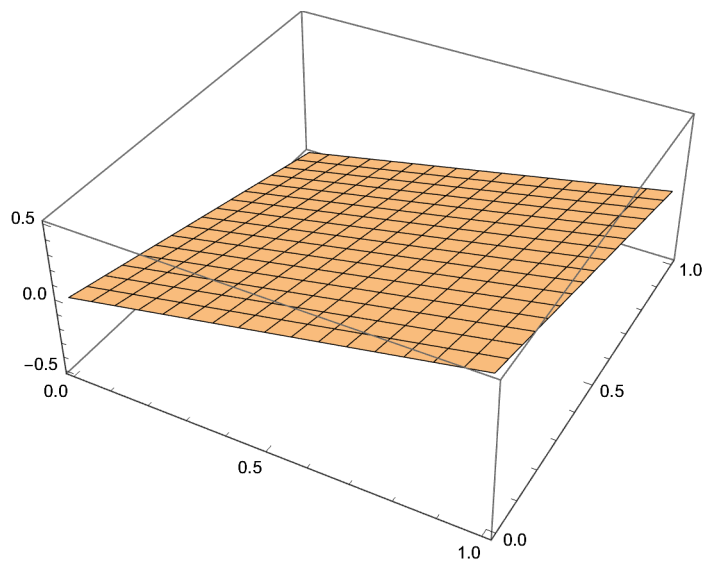
```
f0 = Plot3D[-x + y, {x, 0, 1}, {y, 0, 1}, PlotPoints -> 10]
```




```
f1 = Plot3D[x - y, {x, 0, 1}, {y, 0, 1}, PlotPoints -> 10]
```

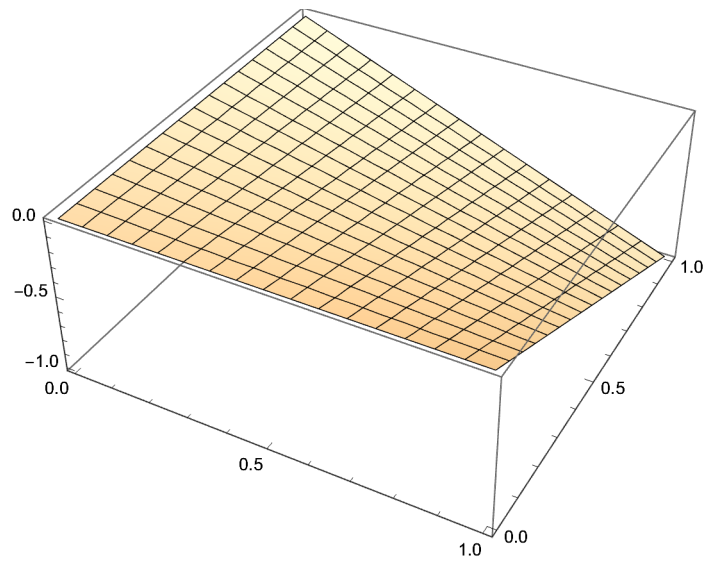


```
f2 = Plot3D[ $\frac{x-y}{2}$ , {x, 0, 1}, {y, 0, 1}, PlotPoints -> 10]
```

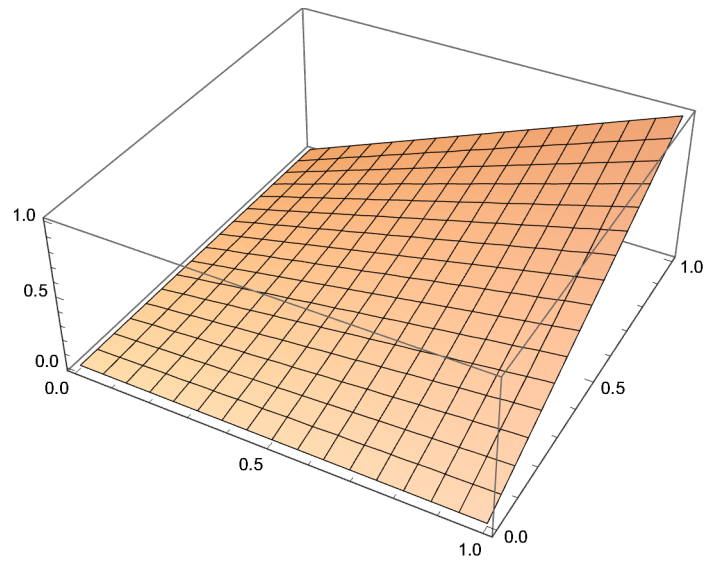


```
g1 = Show[f0, f1, f2];
```

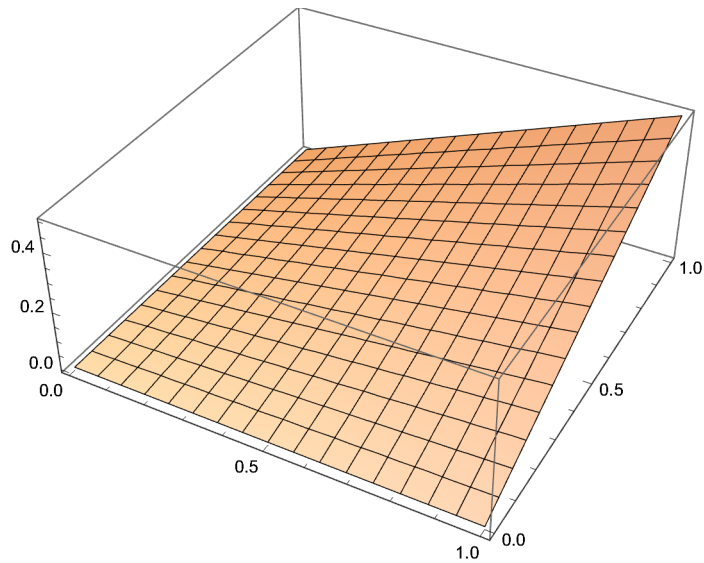
```
h0 = Plot3D[-x y, {x, 0, 1}, {y, 0, 1}, PlotPoints -> 10]
```



```
h1 = Plot3D[x y, {x, 0, 1}, {y, 0, 1}, PlotPoints -> 10]
```

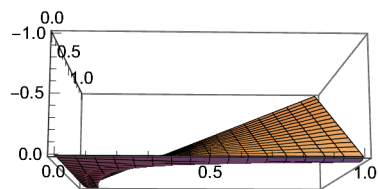
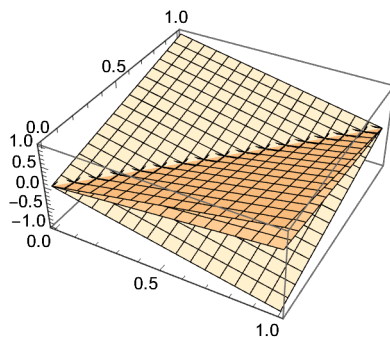


```
h2 = Plot3D[ $\frac{xy}{2}$ , {x, 0, 1}, {y, 0, 1}, PlotPoints -> 10]
```



```
g2 = Show[h0, h1, h2];
```

```
Show[GraphicsGrid[{{g1}, {g2}}]]
```



Q1. Plot the Characteristic curves of the equation: $u_x - u_y = 1$

Solution: The characteristic equations are:

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{du}{1}$$

On taking I = III, we get

```
In[45]:= sol1 = DSolve[u' [x] == 1, u[x], x]
```

```
Out[45]= {{u[x] -> x + C[1]}}
```

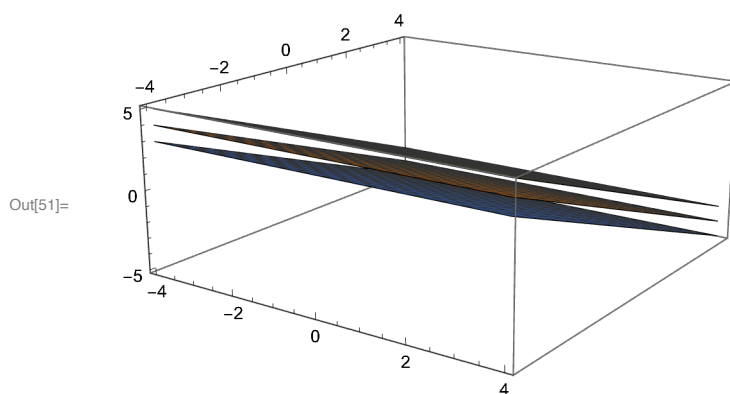
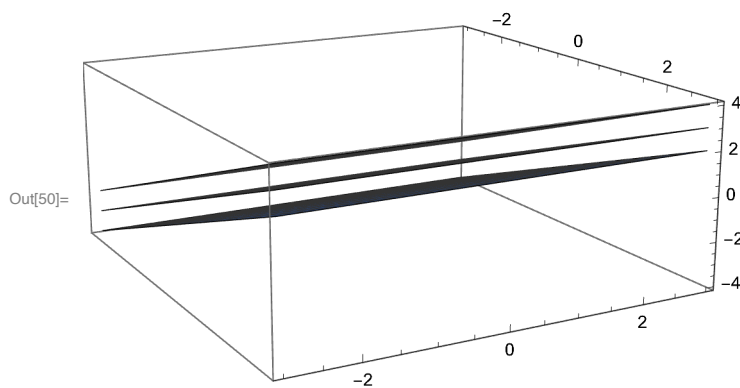
On taking II = III, we get

```
In[47]:= sol2 = DSolve[u' [y] == -1, u[y], y]
```

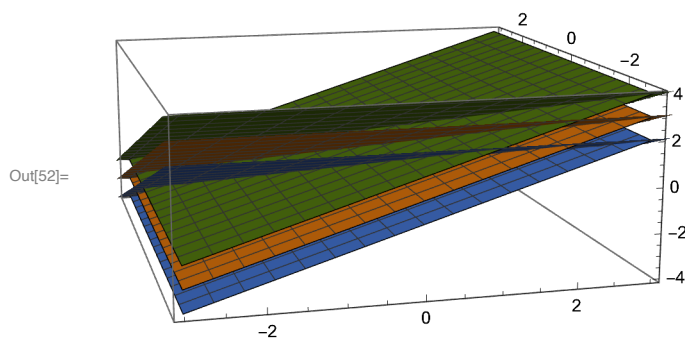
```
Out[47]= {{u[y] -> -y + C[1]}}
```

```
In[50]:= f1 = Plot3D[{x, x - 1, x + 1}, {x, -3, 3}, {y, -3, 3}]
```

```
f2 = Plot3D[{-y, -y - 1, -y + 1}, {x, -4, 4}, {y, -4, 4}]
```



```
In[52]:= Show[f1, f2]
```



Q2. Plot the Characteristic curves of the equation : $u u_x + u_y = 1 \dots \dots$

Solution : The characteristic equations are :

$$\frac{dx}{u} = \frac{dy}{1} = \frac{du}{1}$$

On taking II = III, we get

```
In[53]:= sol1 = DSolve[u' [y] == 1, u[y], y]
```

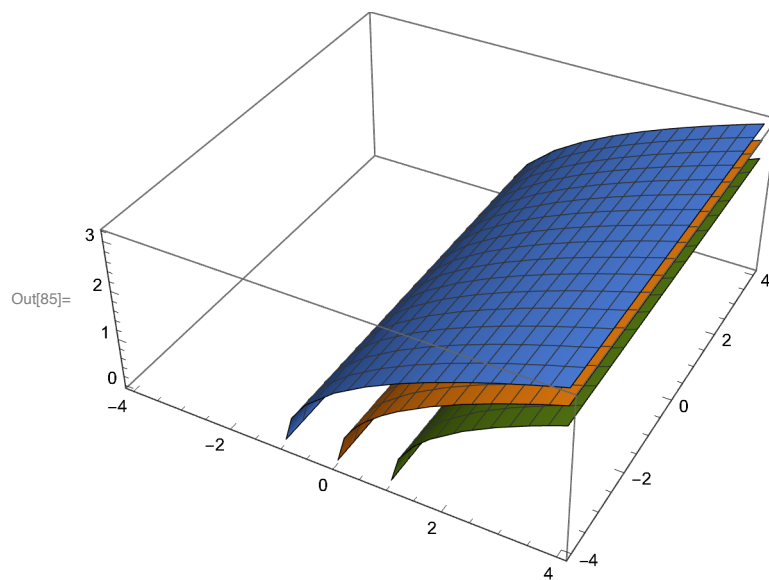
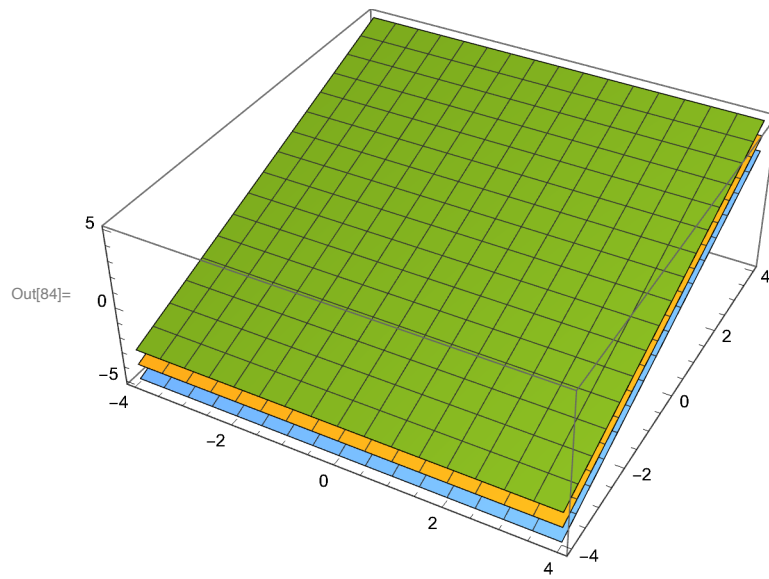
```
Out[53]= {{u[y] -> y + c1}}
```

On taking I= III, we get

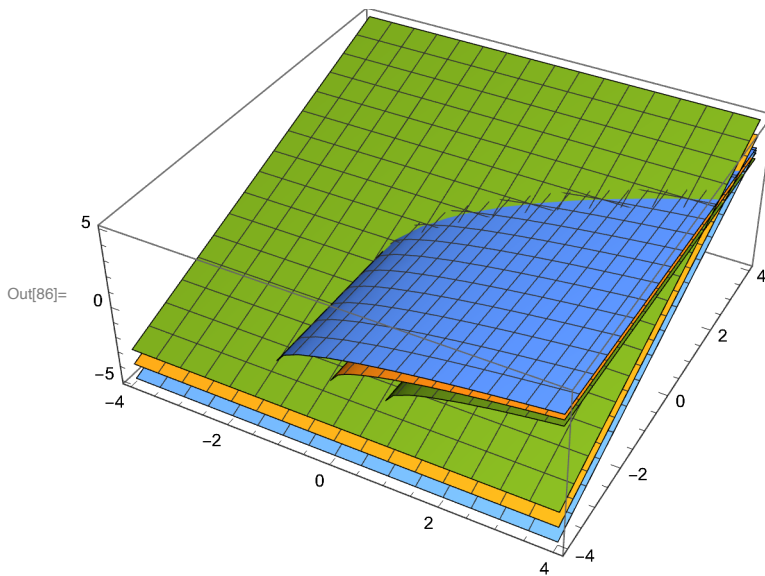
```
In[54]:= sol2 = DSolve[u' [x] * u[x] == 1, u[x], x]
```

```
Out[54]= {{u[x] -> -sqrt(2) sqrt(x + c1)}, {u[x] -> sqrt(2) sqrt(x + c1)}}
```

```
In[84]:= f1 = Plot3D[{y, y - 1, y + 1}, {x, -4, 4}, {y, -4, 4}]
f2 = Plot3D[{ $\sqrt{2} \sqrt{x}$ ,  $\sqrt{2} \sqrt{x+1}$ ,  $\sqrt{2} \sqrt{x-1}$ }, {x, -4, 4}, {y, -4, 4}]
```



In[86]:= Show[f1, f2]



Q3. Plot the Characteristic curves of the equation: $(u - y) u_x + y u_y = x + y$

(1)

Solution: The characteristic equations are:

$$\frac{dx}{u-y} = \frac{dy}{y} = \frac{du}{x+y}$$

On taking I + III = II, we get

In[76]:= Clear[x, y, u, w];

In[77]:= sol1 = DSolve[w'[y] == $\frac{w[y]}{y}$, w[y], y]

Out[77]= {{w[y] → y c₁}}

In[78]:= w = u + x /. sol1

Out[78]= {u + x}

On taking I + II = III, we get

In[81]:= Clear[x, y, u, w];

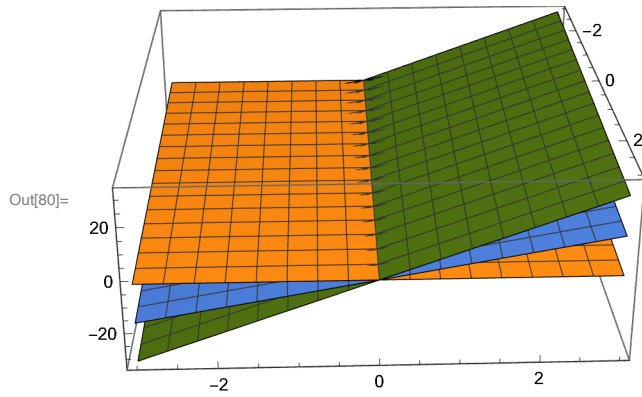
sol2 = DSolve[w'[u] × w[u] == u, w[u], u]

Out[82]= {{w[u] → $-\sqrt{u^2 + 2} c_1$ }, {w[u] → $\sqrt{u^2 + 2} c_1$ }}

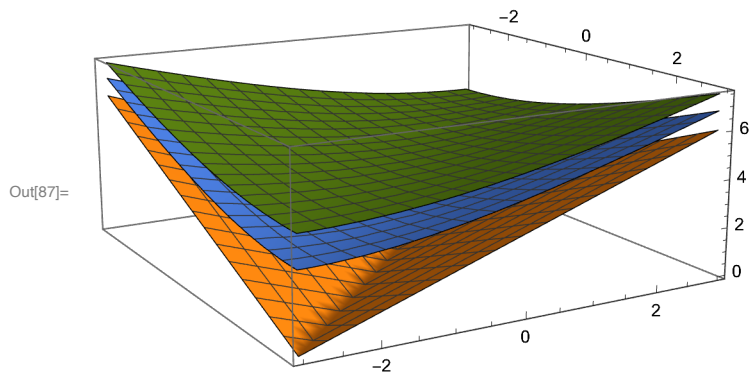
```
In[83]:= w = u + x /. sol2
```

```
Out[83]= {u + x, u + x}
```

```
In[80]:= f1 = Plot3D[{-x, -x + 5 y, -x + 10 y}, {x, -3, 3}, {y, -3, 3}]
```



```
In[87]:= f2 = Plot3D[{Sqrt[(x + y)^2], Sqrt[(x + y)^2 + 10], Sqrt[(x + y)^2 + 20]}, {x, -3, 3}, {y, -3, 3}]
```



Practical4

Solution of vibrating string problem using D'Alembert formula with initial conditions

Problem-:1 Solve the initial value problem

$$u_{tt} = u_{xx}, \quad -\infty < x < \infty, \quad t > 0,$$

$$u(x, 0) = \sin x, \quad -\infty < x < \infty,$$

$$u_t(x, 0) = 0, \quad -\infty < x < \infty$$

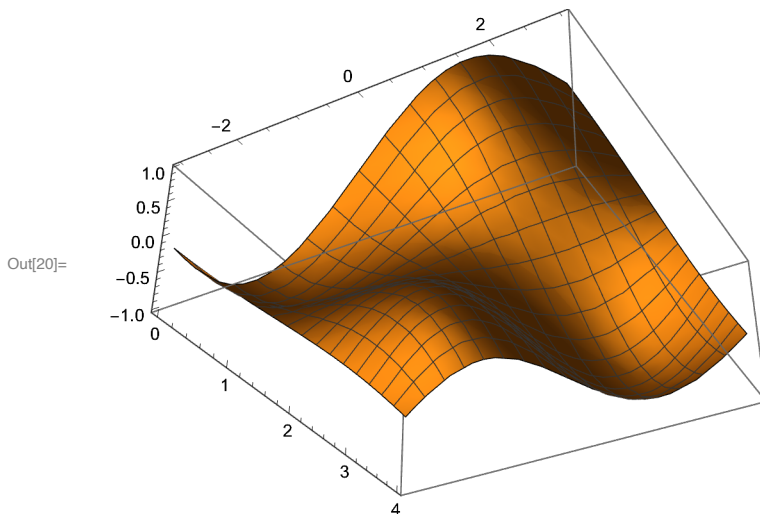
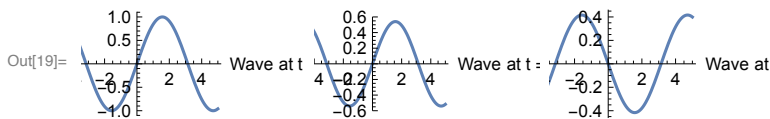
```

In[11]:= c = 1;
f[x_] := Sin[x];
g[x_] := 0;
u[x_, t_] :=  $\frac{1}{2} (f[x + c t] + f[x - c t]) + \frac{1}{2 c} \text{Integrate}[g[s], \{s, x - c t, x + c t\}]$ 
Print[u[x, t]]
h0 = Plot[Evaluate[u[x, 0]], {x, -5, 5},
  PlotRange → All, PlotLegends → "Wave at t =0"];
h1 = Plot[Evaluate[u[x, 1]], {x, -5, 5},
  PlotRange → All, PlotLegends → "Wave at t =1"];
h2 = Plot[Evaluate[u[x, 2]], {x, -5, 5},
  PlotRange → All, PlotLegends → "Wave at t =2"];
Show[GraphicsArray[{h0, h1, h2}]]
Plot3D[u[x, t], {x, -3, 3}, {t, 0, 4}]

$$\frac{1}{2} (-\sin[t - x] + \sin[t + x])$$


```

GraphicsArray : GraphicsArray is obsolete. Switching to GraphicsGrid.



Problem-:2 Solve the initial value problem

$$\begin{aligned}
 u_{tt} &= 4 u_{xx}, & -\infty < x < \infty, \quad t > 0, \\
 u(x, 0) &= e^{-x^2} \sin x, & -\infty < x < \infty, \\
 u_t(x, 0) &= 0, & -\infty < x < \infty
 \end{aligned}$$

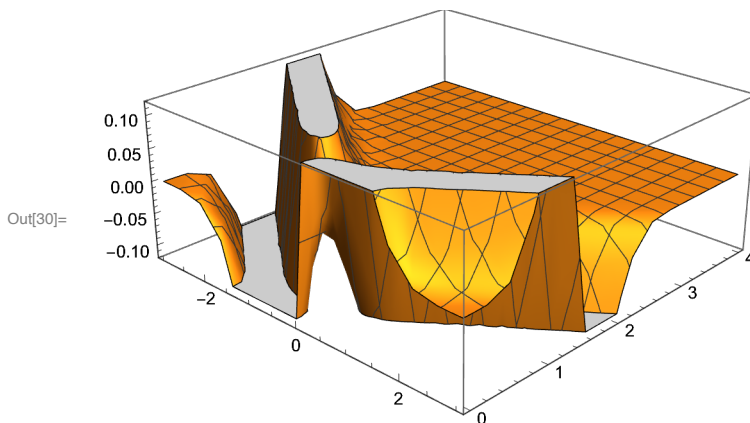
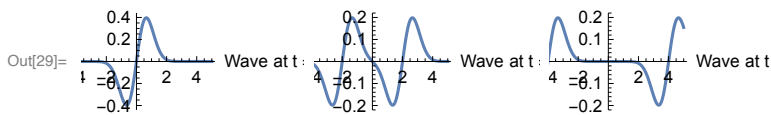
```

In[21]:= c = 2;
f[x_] := Exp[-x^2] Sin[x];
g[x_] := 0;
u[x_, t_] :=  $\frac{1}{2} (f[x + c t] + f[x - c t]) + \frac{1}{2 c} \text{Integrate}[g[s], \{s, x - c t, x + c t\}]$ 
Print[u[x, t]]
h0 = Plot[Evaluate[u[x, 0]], {x, -5, 5},
  PlotRange -> All, PlotLegends -> "Wave at t = 0"];
h1 = Plot[Evaluate[u[x, 1]], {x, -5, 5},
  PlotRange -> All, PlotLegends -> "Wave at t = 1"];
h2 = Plot[Evaluate[u[x, 2]], {x, -5, 5},
  PlotRange -> All, PlotLegends -> "Wave at t = 2"];
Show[GraphicsArray[{h0, h1, h2}]]
Plot3D[u[x, t], {x, -3, 3}, {t, 0, 4}]

$$\frac{1}{2} \left( -e^{-(2t+x)^2} \sin[2t-x] + e^{-(2t-x)^2} \sin[2t+x] \right)$$


```

... GraphicsArray : GraphicsArray is obsolete. Switching to GraphicsGrid.



Problem-:3 Solve the initial value problem

$$u_{tt} = 2 u_{xx}, \quad -\infty < x < \infty, \quad t > 0,$$

$$u(x, 0) = \begin{cases} 0 & x < -1 \\ 1 & -1 \leq x \leq 1, \\ 0 & x > 1 \end{cases}, \quad -\infty < x < \infty,$$

$$u_t(x, 0) = \sin x, \quad -\infty < x < \infty$$

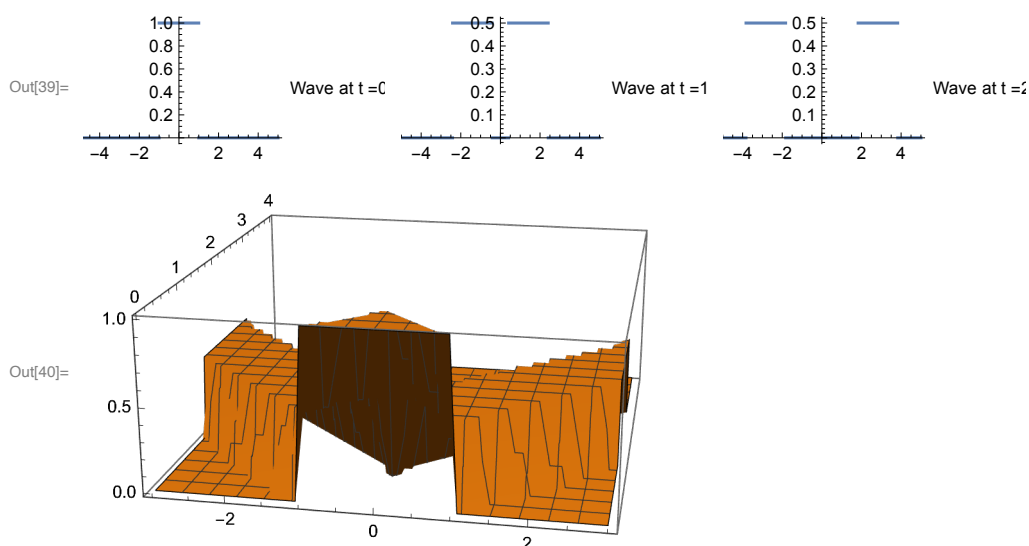
```

In[31]:= c =  $\sqrt{2}$ ;
f[x_] := Piecewise[{{0, x < -1}, {1, -1 ≤ x ≤ 1}, {0, x > 1}}];
g[x_] := 0;
u[x_, t_] :=  $\frac{1}{2}$  (f[x + c t] + f[x - c t]) +  $\frac{1}{2c}$  Integrate[g[s], {s, x - c t, x + c t}]
Print[u[x, t]]
h0 = Plot[Evaluate[u[x, 0]], {x, -5, 5},
  PlotRange → All, PlotLegends → "Wave at t = 0"];
h1 = Plot[Evaluate[u[x, 1]], {x, -5, 5},
  PlotRange → All, PlotLegends → "Wave at t = 1"];
h2 = Plot[Evaluate[u[x, 2]], {x, -5, 5},
  PlotRange → All, PlotLegends → "Wave at t = 2"];
Show[GraphicsArray[{h0, h1, h2}]]
Plot3D[u[x, t], {x, -3, 3}, {t, 0, 4}]

$$\frac{1}{2} \left( \begin{cases} 0 & -\sqrt{2} t + x < -1 \\ 1 & -1 \leq -\sqrt{2} t + x \leq 1 \\ 0 & \text{True} \end{cases} + \begin{cases} 0 & \sqrt{2} t + x < -1 \\ 1 & -1 \leq \sqrt{2} t + x \leq 1 \\ 0 & \text{True} \end{cases} \right)$$


```

GraphicsArray : GraphicsArray is obsolete. Switching to GraphicsGrid.



Problem-:4 Solve the initial value problem

$$\begin{aligned}
 u_{tt} &= 2 u_{xx}, & -\infty < x < \infty, \quad t > 0, \\
 u(x, 0) &= \sin x, & -\infty < x < \infty, \\
 u_t(x, 0) &= \cos x, & -\infty < x < \infty
 \end{aligned}$$

Problem-:5 Solve the initial value problem

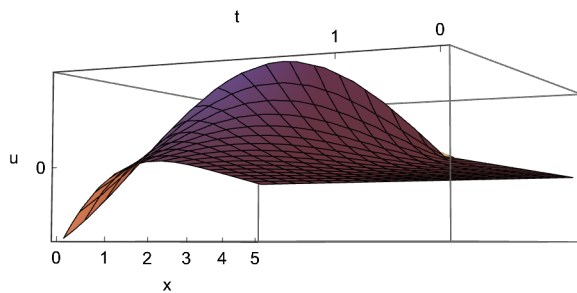
$$\begin{aligned}
 u_{tt} &= \pi u_{xx}, & -\infty < x < \infty, \quad t > 0, \\
 u(x, 0) &= 0, & -\infty < x < \infty, \\
 u_t(x, 0) &= e^{-x^2}, & -\infty < x < \infty
 \end{aligned}$$

Practical : –(5)

Solution of Heat Equations

Problem-:1(a) $u_t - u_{xx} = 0,$ $0 < x < 5,$ $t > 0,$
 $u(x, 0) = 0,$ $0 \leq x \leq 5,$
 $u(0, t) = \sin(t)$ $t \geq 0$
 $u(5, t) = 0,$ $t \geq 0.$

```
eqn1a = {∂tu[x, t] - ∂x,xu[x, t] == 0,  
u[x, 0] == 0, u[0, t] == Sin[t], u[5, t] == 0}  
sol1a = u[x, t] /. NDSolve[eqn1a, u[x, t],  
{x, 0, 5}, {t, 0, 10}, PrecisionGoal → 3] [[1]]  
Plot3D[sol1a, {x, 0, 5}, {t, 0, 4}, AxesLabel → {"x", "t", "u"},  
Ticks → {{0, 1, 2, 3, 4, 5}, {0, 1}, {-3, 0}}]  
{u(0,1)[x, t] - u(2,0)[x, t] == 0, u[x, 0] == 0, u[0, t] == Sin[t], u[5, t] == 0}  
InterpolatingFunction[{{0., 5.}, {0., 10.}}, <>][x, t]
```

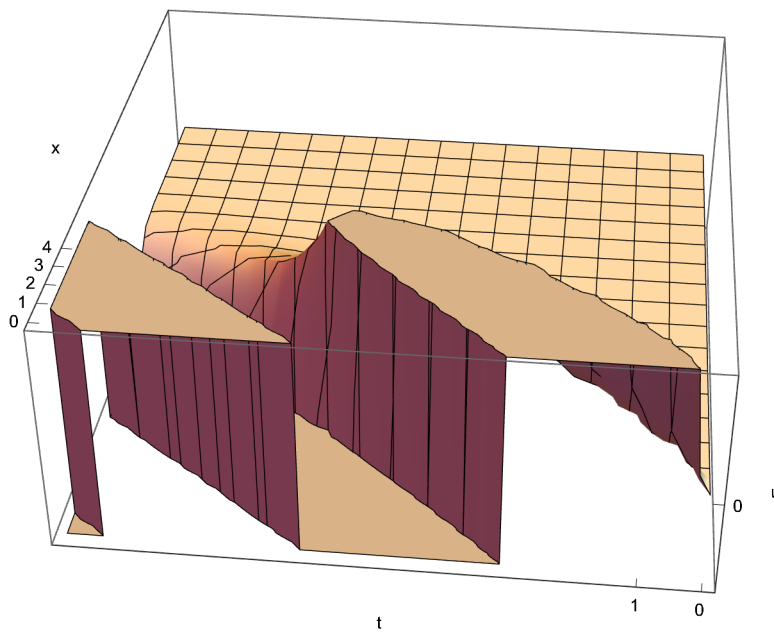


Problem-:1(b) $u_t - u_{xx} = 0,$ $0 < x < 20,$ $t > 0,$
 $u(x, 0) = 0,$ $0 \leq x \leq 20,$
 $u(0, t) = t^2 * \sin(t)$ $t \geq 0$
 $u(20, t) = 0,$ $t \geq 0.$

```

eqn1b = { $\partial_t u[x, t] - \partial_{x,x} u[x, t] == 0$ ,
  u[x, 0] == 0, u[0, t] ==  $t^2 * \text{Sin}[t]$ , u[20, t] == 0}
sol1b = u[x, t] /. NDSolve[eqn1b, u[x, t],
  {x, 0, 20}, {t, 0, 10}, PrecisionGoal -> 3][[1]]
Plot3D[sol1b, {x, 0, 20}, {t, 0, 10}, AxesLabel -> {"x", "t", "u"},
  Ticks -> {{0, 1, 2, 3, 4}, {0, 1}, {-3, 0}}]
{u(0,1)[x, t] - u(2,0)[x, t] == 0, u[x, 0] == 0, u[0, t] ==  $t^2 \text{Sin}[t]$ , u[20, t] == 0}
InterpolatingFunction[{{0., 20.}, {0., 10.}}, <>][x, t]

```

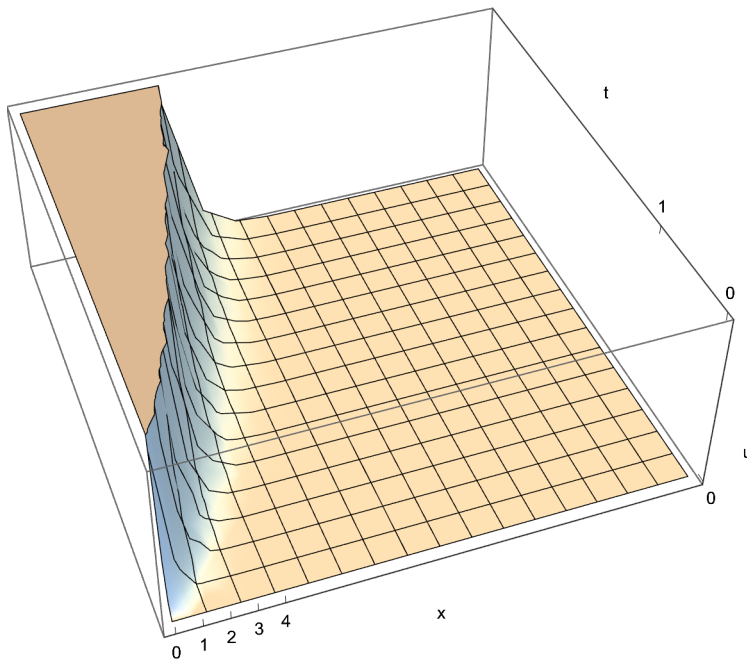


Problem-:1(c) $u_t - u_{xx} = 0$, $0 < x < 20$, $t > 0$,
 $u(x, 0) = 0$, $0 \leq x \leq 20$,
 $u(0, t) = t^2$, $t \geq 0$
 $u(20, t) = 0$, $t \geq 0$.

```

eqn1c =
  { $\partial_t u[x, t] - \partial_{x,x} u[x, t] == 0$ ,  $u[x, 0] == 0$ ,  $u[0, t] == t^2$ ,  $u[20, t] == 0$ }
sol1c = u[x, t] /. NDSolve[eqn1c, u[x, t],
  {x, 0, 20}, {t, 0, 10}, PrecisionGoal → 3][[1]]
Plot3D[sol1c, {x, 0, 20}, {t, 0, 4}, AxesLabel → {"x", "t", "u"},
  Ticks → {{0, 1, 2, 3, 4}, {0, 1}, {-3, 0}}]
{ $u^{(0,1)}[x, t] - u^{(2,0)}[x, t] == 0$ ,  $u[x, 0] == 0$ ,  $u[0, t] == t^2$ ,  $u[20, t] == 0$ }
InterpolatingFunction[{{0., 20.}, {0., 10.}}, <>][x, t]

```



*******End*******

Problem-:1(d) $u_t = u_{xx}$, $0 < x < \pi$, $t > 0$,
 $u(x, 0) = \sin^2 x$, $0 \leq x \leq \pi$,
 $u(0, t) = 0$, $t \geq 0$
 $u(\pi, t) = 0$, $t \geq 0$.

Practical - 6

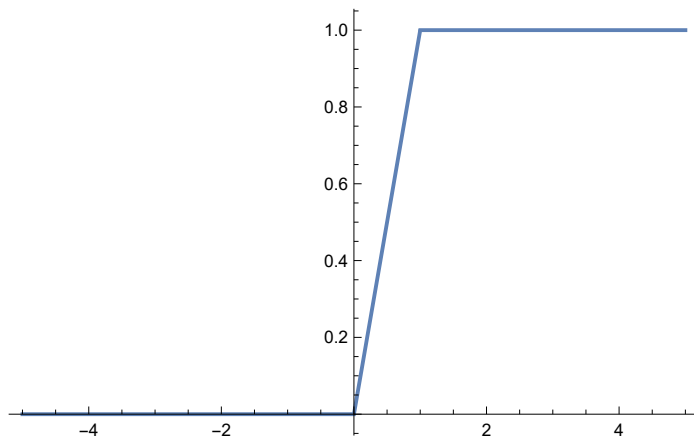
```
In[*]:= eqn = D[u[x, t], t] + 2 u[x, t] × D[u[x, t], x] == 0;
f[x_] := Piecewise[{{0, x ≤ 0}, {x, 0 < x < 1}, {1, 1 ≤ x}}]
DSolve[eqn, f[x_], u[x, t], {x, t}]
Plot[f[x], {x, -5, 5}]
```

DSolve: {x, t} cannot be used as a variable.

Out[*]=

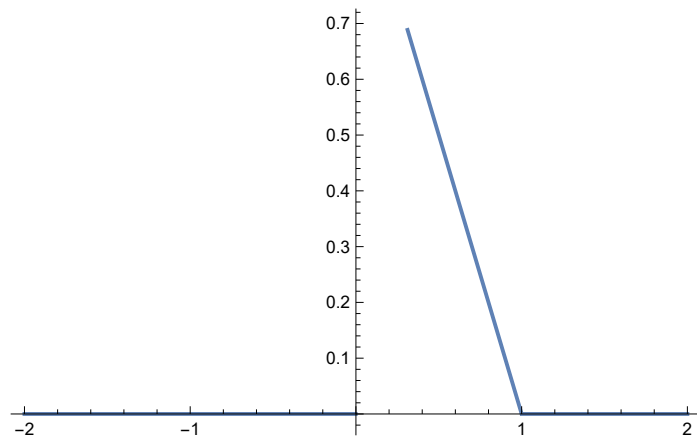
DSolve $\left[u^{(0,1)}[x, t] + 2 u[x, t] u^{(1,0)}[x, t] == 0, \begin{cases} 0 & x_- \leq 0 \\ x_- & 0 < x_- < 1 \\ 1 & 1 \leq x_- \\ 0 & \text{True} \end{cases}, u[x, t], \{x, t\} \right]$

Out[*]=

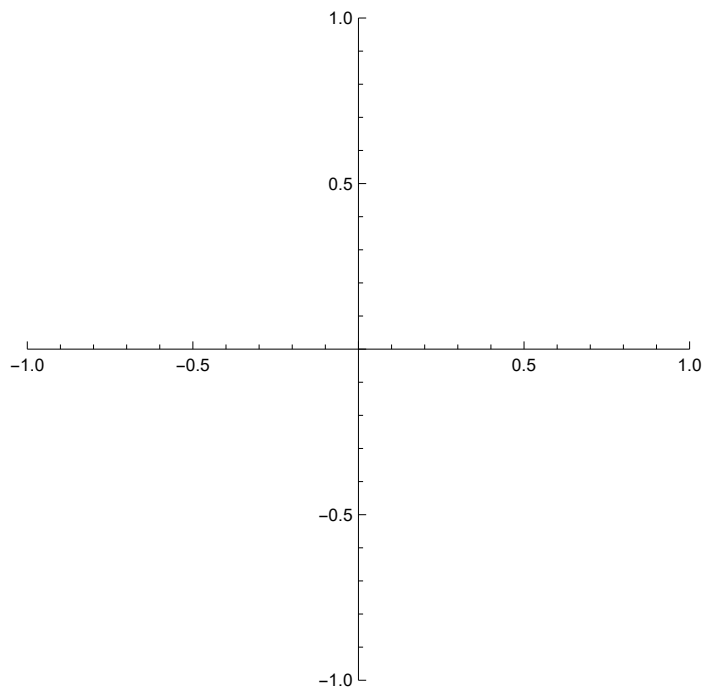


```
In[*]:= f[x_] := Piecewise[{{0, x ≤ 0}, {1 - x, 0 < x < 1}, {0, x ≥ 1}}];
Plot[f[x], {x, -2, 2}]
y[x_, t_] := x + t * f[x];
ParametricPlot[{{y[-1.5, t], t}, {y[-1, t], t},
  {y[0, t], t}, {y[0.75, t], t}, {y[1.5, t], y[2, t], t}}, {t, 0, 3}]
h1 = ParametricPlot[{y[x, 0], f[x]}, {x, -1, 2}, PlotRange → {-1, 2}];
h2 = ParametricPlot[{y[x, 0.5], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
h3 = ParametricPlot[{y[x, 1], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
h4 = ParametricPlot[{y[x, 1.5], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
ParametricPlot3D[{t f[x] + x, t, f[x]}, {x, -2, 2}, {t, 0, 3}]
Show[h1, h2, h3, h4]
```

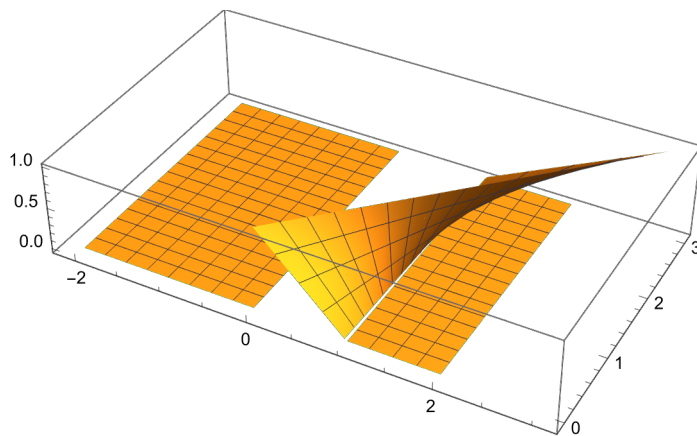

Out[*]=



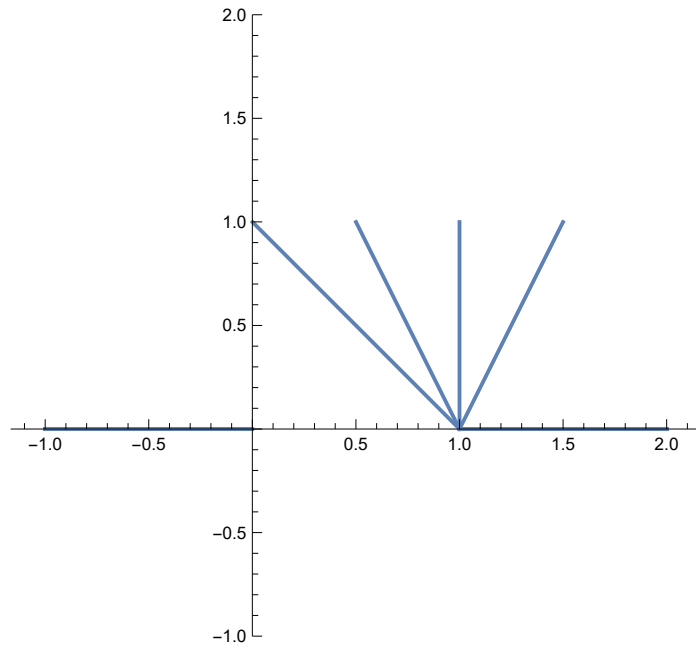
Out[*]=



Out[*]=



Out[8]=

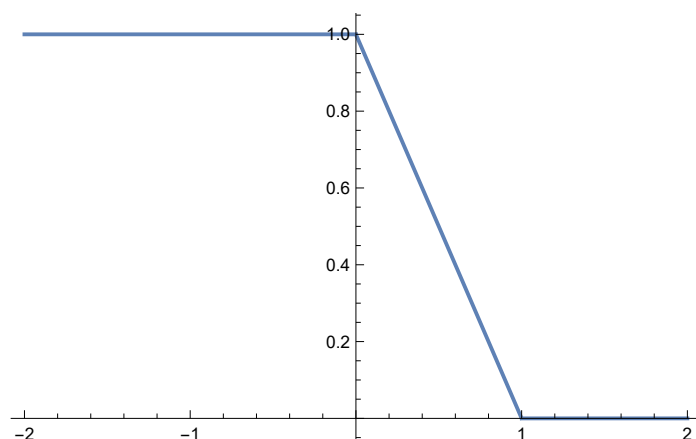


```

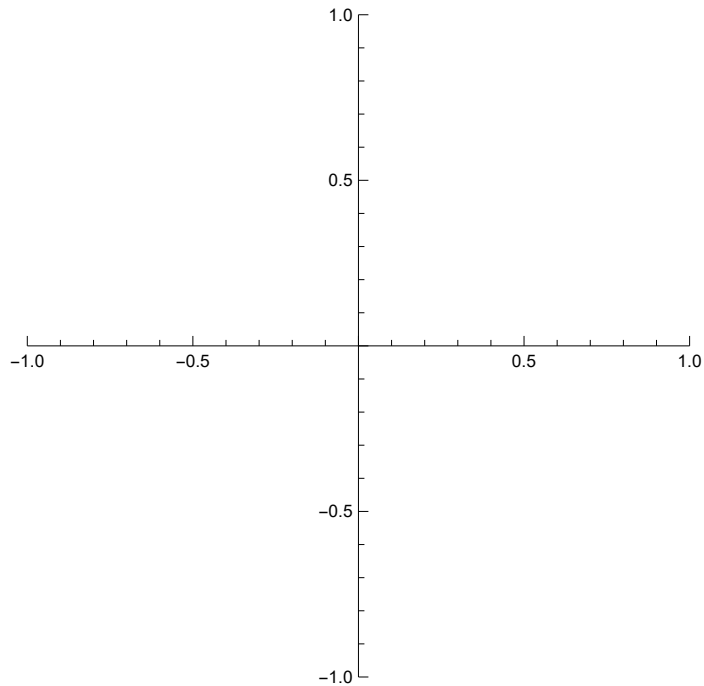
In[9]:= f[x_] := Piecewise[{{1, x ≤ 0}, {1 - x, 0 < x < 1}, {0, x ≥ 1}}]
Plot[f[x], {x, -2, 2}]
y[x_, t_] := x + t * f[x];
ParametricPlot[{{y[-1.5, t], t}, {y[-1, t], t},
  {y[0, t], t}, {y[0.75, t], t}, {y[1.5, t], y[2, t], t}}, {t, 0, 3}]
h1 = ParametricPlot[{y[x, 0], f[x]}, {x, -1, 2}, PlotRange → {-1, 2}];
h2 = ParametricPlot[{y[x, 0.5], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
h3 = ParametricPlot[{y[x, 1], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
h4 = ParametricPlot[{y[x, 1.5], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
ParametricPlot3D[{t f[x] + x, t, f[x]}, {x, -2, 2}, {t, 0, 3}]

```

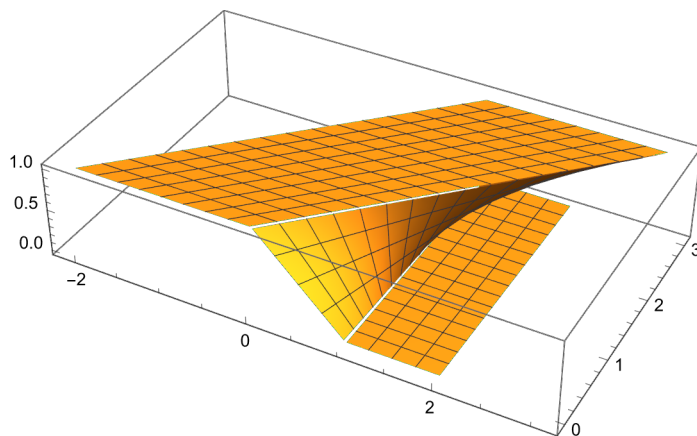
Out[9]=



Out[8]=



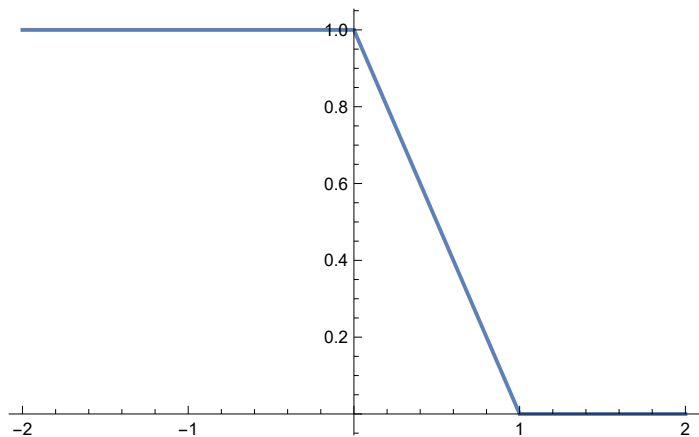
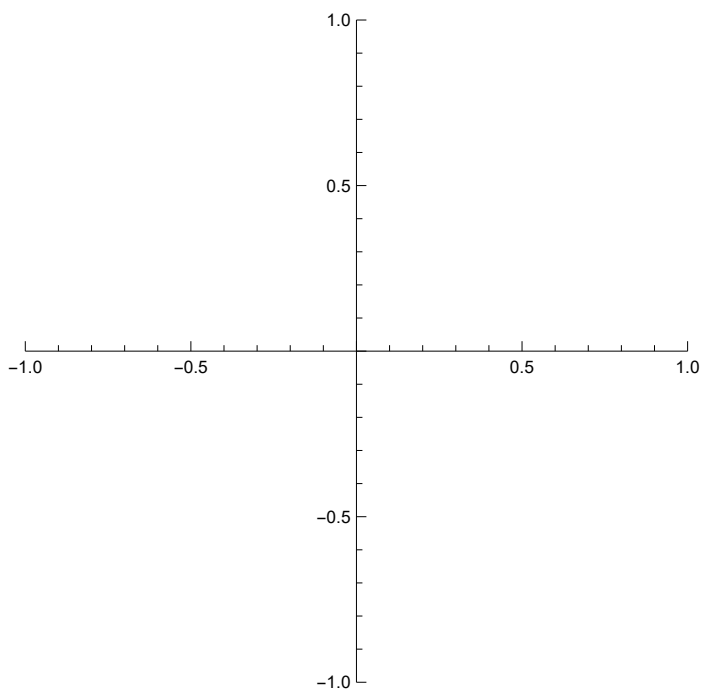
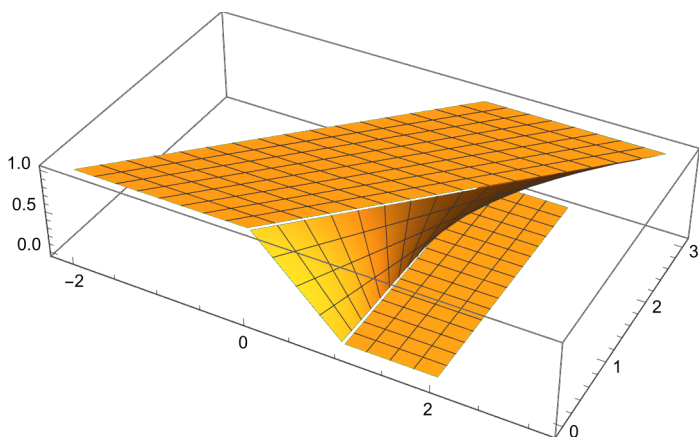
Out[9]=



```

In[10]:= f[x_] := Piecewise[{{1, x ≤ 0}, {1 - x, 0 < x < 1}, {0, x ≥ 1}}]
Plot[f[x], {x, -2, 2}]
y[x_, t_] := x + t * (1 - 2 f[x]);
ParametricPlot[{{y[-1.5, t], t}, {y[-1, t], t},
  {y[0, t], t}, {y[0.75, t], t}, {y[1.5, t], y[2, t], t}}, {t, 0, 3}]
h1 = ParametricPlot[{y[x, 0], f[x]}, {x, -1, 2}, PlotRange → {-1, 2}];
h2 = ParametricPlot[{y[x, 0.5], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
h3 = ParametricPlot[{y[x, 1], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
h4 = ParametricPlot[{y[x, 1.5], f[x]}, {x, -1, 2}, PlotRange → {-1, 3}];
ParametricPlot3D[{t f[x] + x, t, f[x]}, {x, -2, 2}, {t, 0, 3}]

```

Out[*n*]=Out[*n*]=Out[*n*]=

Practical 7

Solution of traffic flow problem with given initial conditions, and plotting of the characteristic base curves and the traffic density function.

Q1. Find the traffic density $u(x,t)$ satisfying

$$\frac{\partial u}{\partial t} + x \sin t \frac{\partial u}{\partial x} = 0.$$

with the initial condition $u(x,0) = 1 + \frac{1}{1+x^2}$.

Solution: The characteristic equations are:

$$\frac{dt}{1} = \frac{dx}{x \sin t} = \frac{du}{0}$$

On taking $I = II$, we get the characteristic base curves

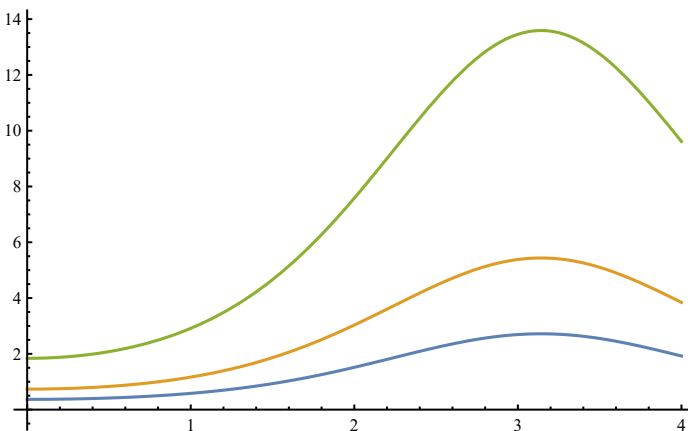
```
In[*]:= sol1 = DSolve[x' [t] == x [t] Sin[t], x[t], t]
```

```
Out[*]=
```

$$\left\{ \left\{ x[t] \rightarrow e^{-\cos[t]} C_1 \right\} \right\}$$

```
In[*]:= Plot[{e^{-Cos[t]}, 2 e^{-Cos[t]}, 5 e^{-Cos[t]}}, {t, 0, 4}]
```

```
Out[*]=
```



The traffic density function is given by

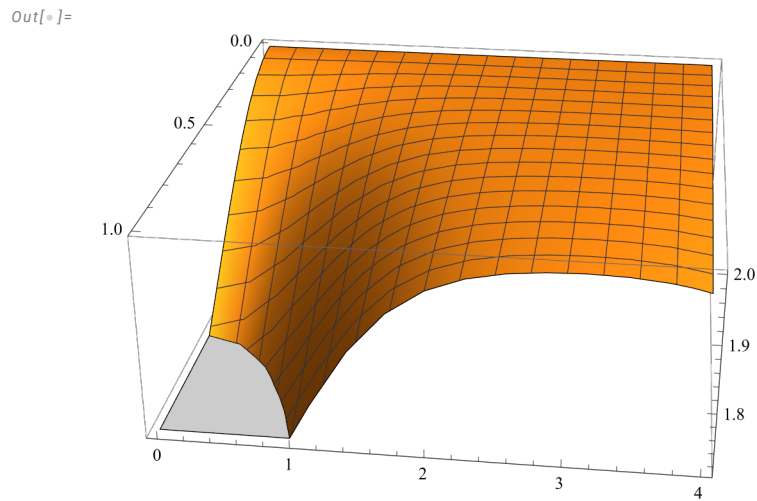
```
In[*]:= eqn2 = D[u[x, t], t] + x Sin[t] D[u[x, t], x] == 0;
sol2 = DSolve[{eqn2, u[x, 0] == 1 +  $\frac{1}{1+x^2}$ }, u[x, t], {x, t}]
```

... Solve: Inverse functions are being used by Solve, so some solutions may not be found; use Reduce for complete solution information.

```
Out[*]:=
```

$$\left\{ \left\{ u[x, t] \rightarrow \frac{2 e^{2 + e^{2 \cos[t]} x^2}}{e^{2 + e^{2 \cos[t]} x^2}} \right\} \right\}$$

```
In[*]:= Plot3D[u[x, t] /. sol2, {x, 0, 1}, {t, 0, 4}]
```



Q2. Find the traffic density $u(x, t)$ satisfying

$$\frac{\partial u}{\partial t} + 2 \frac{\partial u}{\partial x} = 0.$$

with the initial condition $u(x, 0) = \begin{cases} 0 & x < 0 \\ 4x & 0 < x < 1 \\ 0 & x > 1 \end{cases}$,

Solution: The characteristic equations are:

$$\frac{dt}{1} = \frac{dx}{2} = \frac{du}{0}$$

On taking I = III, we get

```
In[*]:= ClearAll;
```

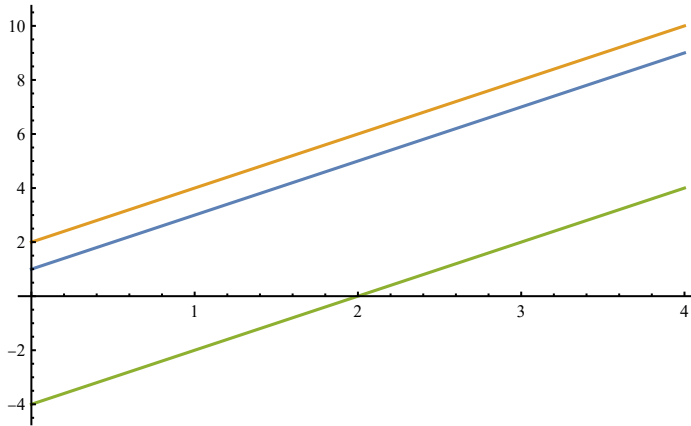
```
In[*]:= sol2 = DSolve[x'[t] == 2, x[t], t]
```

```
Out[*]:=
```

$$\{ \{ x[t] \rightarrow 2t + c_1 \} \}$$

```
In[*]:= Plot[{2 t + 1, 2 t + 2, 2 t - 4}, {t, 0, 4}]
```

```
Out[*]:=
```



```
In[*]:= f[x_] := Piecewise[{{0, x < 0}, {4 x, 0 < x < 1}, {0, x > 1}}];
```

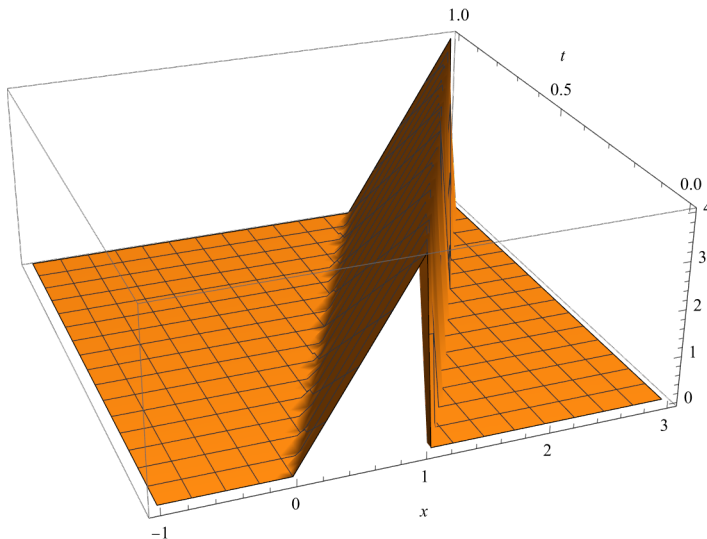
```
In[*]:= eqn2 = D[u[x, t], t] + 2 D[u[x, t], x] == 0;  
sol2 = DSolve[{eqn2, u[x, 0] == f[x]}, u[x, t], {x, t}]
```

```
Out[*]:=
```

$$\left\{ \left\{ u[x, t] \rightarrow \begin{cases} 0 & -2 \left(t - \frac{x}{2} \right) < 0 \\ -4 (2 t - x) & 0 < -2 \left(t - \frac{x}{2} \right) < 1 \\ 0 & \text{True} \end{cases} \right\} \right\}$$

```
In[*]:= Plot3D[u[x, t] /. sol2, {x, -1, 3}, {t, 0, 1}, AxesLabel -> Automatic]
```

```
Out[*]:=
```



Q3. Find the traffic density $u(x, t)$ satisfying

$$\frac{\partial u}{\partial t} + (1 - 2u) \frac{\partial u}{\partial x} = 0.$$

with the initial condition $u(x,0) = \begin{cases} 4, & x < 0 \\ 3 & x > 0 \end{cases}$

Solution: The characteristic equations are:

$$\frac{dt}{1} = \frac{dx}{1-2u} = \frac{du}{0}$$

On taking I=III, we get

```
In[ ]:= sol1 = DSolve[u'[t] == 0, u[t], t]
```

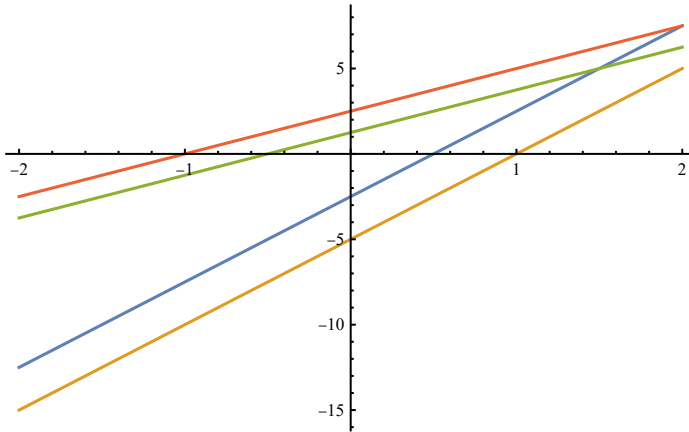
```
Out[ ]:= {{u[t] -> c1}}
```

On Replacing u by c, and taking I=II, we get

```
In[ ]:= ClearAll;
```

```
In[ ]:= Plot[{5 (x - .5), 5 (x - 1), 5/2 (x + .5), 5/2 (x + 1)}, {x, -2, 2}]
```

```
Out[ ]:=
```



```
In[ ]:= f[x_] := Piecewise[{{4, x < 0}, {3, x > 0}}];
eqn3 = D[u[x, t], t] + (1 - 2 u[x, t]) D[u[x, t], x] == 0;
sol3 = NDSolve[{eqn3, u[x, 0] == f[x]}, u[x, t], {x, t}]
```

... **NDSolve:** There are more dependent variables, $\{u[x, 0], u[x, t], u^{(0,1)}[x, t]\}$, than equations, so the system is underdetermined.

```
Out[ ]:=
```

$$\text{NDSolve}\left[\left\{u^{(0,1)}[x, t] + (1 - 2 u[x, t]) u^{(1,0)}[x, t] == 0, u[x, 0] == \begin{cases} 4 & x < 0 \\ 3 & x > 0 \\ 0 & \text{True} \end{cases}\right\}, u[x, t], \{x, t\}\right]$$

```
eqn3= D[u[x,t],t]+(1-2 u[x,t]) D[u[x,t],x]==0;
sol3= NDSolve[{eqn3, u[x,0]== f[x]}, u[x,t], {x,t}]
```

```
Out[ ]:=
```

$$\text{DSolve}\left[\left\{u^{(0,1)}[x, t] + (1 - 2 u[x, t]) u^{(1,0)}[x, t] == 0, u[x, 0] == \begin{cases} 4 & x < 0 \\ 3 & x > 0 \\ 0 & \text{True} \end{cases}\right\}, u[x, t], \{x, t\}\right]$$


```
In[*]:= eqn3 = D[u[x, t], t] + (1 - 2 u[x, t]) D[u[x, t], x] == 0
sol3 = DSolve[{D[u[x, t], t] + 1/5 D[u[x, t], x] == 0, u[x, 0] == f[x]}, u[x, t], {x, t}]
```

```
Out[*]:=
```

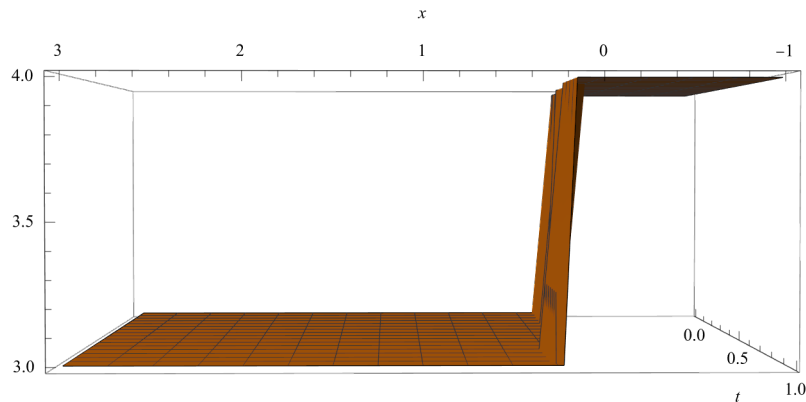
$$u^{(0,1)}[x, t] + (1 - 2 u[x, t]) u^{(1,0)}[x, t] == 0$$

```
Out[*]:=
```

$$\left\{ \left\{ u[x, t] \rightarrow \begin{cases} 4 & \frac{1}{5}(-t + 5x) < 0 \\ 3 & \frac{1}{5}(-t + 5x) > 0 \end{cases} \right\} \right\}$$

```
In[*]:= Plot3D[u[x, t] /. sol3, {x, -1, 3}, {t, 0, 1}, AxesLabel -> Automatic]
```

```
Out[*]:=
```



Q3. Find the traffic density $u(x, t)$ satisfying

$$\frac{\partial u}{\partial t} + 2u \frac{\partial u}{\partial x} = 0.$$

$$\text{with the initial condition } u(x, 0) = \begin{cases} 1 & x < 0 \\ 1 - x & 0 < x < 1 \\ 0 & 1 \leq x \end{cases}$$

Solution : The characteristic equations are :

$$\frac{dt}{1} = \frac{dx}{2u} = \frac{du}{0}$$

On taking I = III, we get

```
In[*]:= sol1 = DSolve[u'[t] == 0, u[t], t]
```

```
Out[*]:=
```

$$\{ \{ u[t] \rightarrow c_1 \} \}$$

On Replacing u by c, and taking I = II, we get

```
In[*]:= sol2 = DSolve[t'[x] == 1/(2c), t[x], x]
```

```
Out[*]:=
```

$$\left\{ \left\{ t[x] \rightarrow \frac{x}{2c} + c_1 \right\} \right\}$$

For $x < 0$, we have $c = u(x, 0) = 1$. Thus the characteristic curves are

```
In[ ]:= h1 = Plot[{1/2 (x + 3), 1/2 (x + 4.5), 1/2 (x + 2)}, {x, -2, 0}];
```

For $0 < x < 1$, we have $c = u(x, 0) = 1 - x$. Thus the characteristic curves are

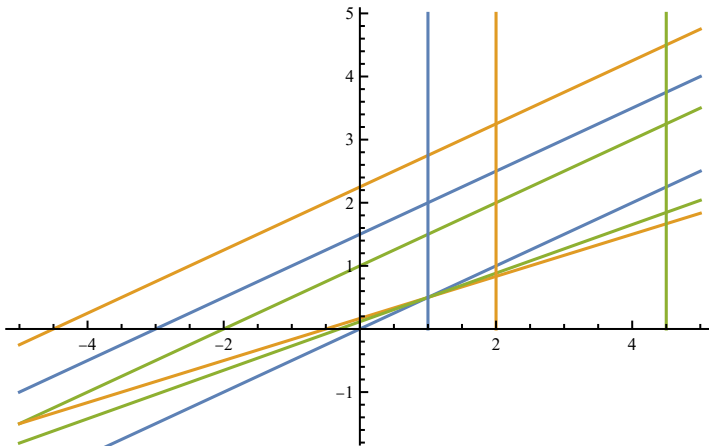
```
In[ ]:= h2 = Plot[{1/2 (x), 1/2 (x + .5), 1/2 (x + 0.3)}, {x, 0, 1}];
```

For $x > 1$, we have $c = u(x, 0) = 0$. Thus the characteristic curves are

```
In[ ]:= h3 = ParametricPlot[{1, t}, {2, t}, {4.5, t}, {t, 0, 5}];
```

```
In[ ]:= Show[Plot[{1/2 (x + 3), 1/2 (x + 4.5), 1/2 (x + 2)}, {x, -5, 5}],
  Plot[{1/2 (x), 1/2 (x + .5), 1/2 (x + 0.3)}, {x, -5, 5}],
  ParametricPlot[{1, t}, {2, t}, {4.5, t}, {t, 0, 5}]]
```

Out[]:=



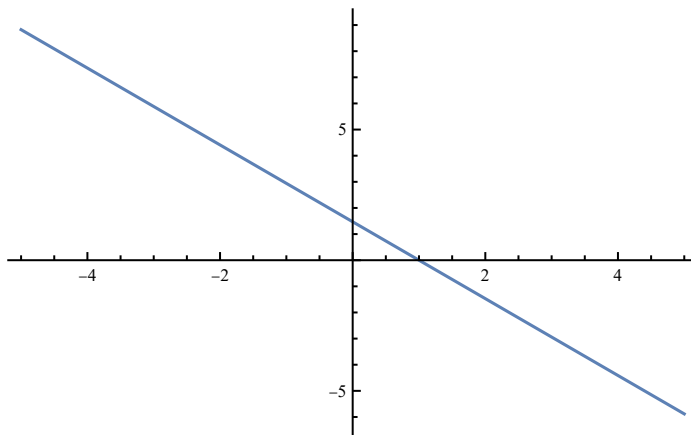
Plotting Density Function for $0 < x < 1$

$$u(x,t) = \frac{1-x}{1-2t}$$

```
In[ ]:= u[x_, t_] := (1 - x) / (1 - 2 t);
```

```
In[*]:= Plot[ $\frac{1-x}{1-2(0.16)}$ , {x, -5, 5}]
```

```
Out[*]=
```



Q3. Find the traffic density $u(x, t)$ satisfying

$$\frac{\partial u}{\partial t} + c(1 - 2u) \frac{\partial u}{\partial x} = 0.$$

$$\text{with the initial condition } u(x, 0) = \begin{cases} \frac{1}{3} & x < 0 \\ \frac{1}{3} + \frac{5}{12}x & 0 < x < 1 \\ \frac{3}{4} & 1 \leq x \end{cases}$$

Solution : The characteristic equations are :

$$\frac{dt}{1} = \frac{dx}{c(1 - 2u)} = \frac{du}{0}$$

On taking I = III, we get

```
In[*]:= sol1 = DSolve[u'[t] == 0, u[t], t]
```

```
Out[*]=
```

```
{{u[t] -> c1}}
```

On Replacing u by k, and taking I = II, we get

```
In[*]:= sol2 = DSolve[x'[t] == c(1 - 2k), x[t], t]
```

```
Out[*]=
```

```
{{x[t] -> c t - 2 c k t + c1}}
```

For $x < 0$, we have $k = u(x, 0) = 1/3$ which gives $ct = 3(x - x_0)$

```
h1 = Plot[{3 x, 3 (x + 1), 3 (x + 2)}, {x, -2, 0}];
```

For $0 < x < 1$, we have $c = u(x, 0) = \frac{1}{3} + \frac{5}{12}x$ which gives $ct = \frac{6(x - x_0)}{5(2/5 - x_0)}$

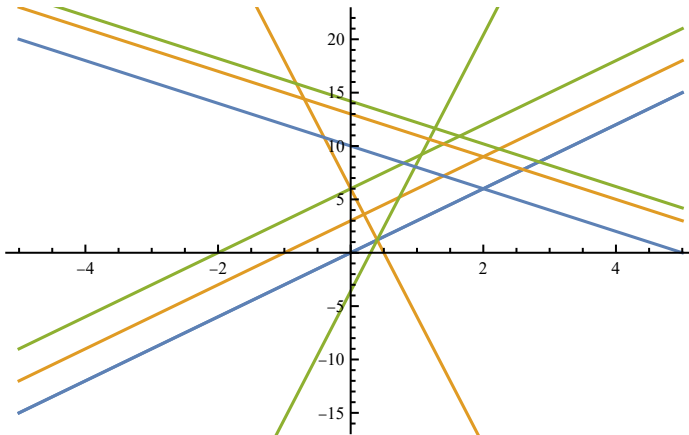
```
In[*]:= h2 = Plot[{ $\frac{6(x)}{5(2/5)}$ ,  $\frac{6(x-0.5)}{5(2/5-0.5)}$ ,  $\frac{6(x-0.3)}{5(2/5-0.3)}$ }, {x, 0, 1}];
```

For $x > 1$, we have $c = u(x, 0) = \frac{3}{4}$ which gives $ct = -2(x - x_0)$

```
In[ ]:= h3 = Plot[{-2 (x - 5), -2 (x - 6.5), -2 (x - 7.1)}, {x, 1, 4}];
```

```
In[ ]:= Show[Plot[{3 x, 3 (x + 1), 3 (x + 2)}, {x, -5, 5}],
  Plot[{ $\frac{6 (x)}{5 (2/5)}$ ,  $\frac{6 (x - 0.5)}{5 (2/5 - 0.5)}$ ,  $\frac{6 (x - 0.3)}{5 (2/5 - 0.3)}$ }, {x, -5, 5}],
  Plot[{-2 (x - 5), -2 (x - 6.5), -2 (x - 7.1)}, {x, -5, 5}]]
```

Out[]:=



Plotting Density Function for $0 < x < 1$

$$u(x, t) = \frac{1}{3} + \frac{5}{12} \frac{6x - 2ct}{6 - 5ct}$$

```
In[ ]:= Plot[ $\frac{1}{3} + \frac{5}{12} \frac{6x}{6}$ , {x, -3, 3}]
```

Out[]:=

